

meant an investment of millions of dollars and perhaps more than one supercomputer.

Folding@home project. This project is somewhat similar to SETI@home project and is administered by Pande Group, a nonprofit institution in Stanford University's chemistry department. It involves studying proteins. The research includes the way proteins take certain shapes, called folds, and how that relates to what proteins do. Scientists believe that protein misfolding could be the cause of diseases like Parkinson's or Alzheimer's.

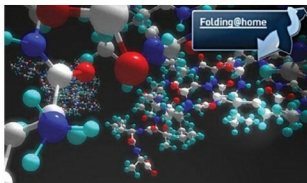
Hence, there is a possibility that by studying proteins, Pande Group may unearth novel ways to treat or even cure such diseases.

World Community Grid (WCG) project. This project is an initiative to create the world's largest public computing grid to process scientific research projects that benefit humanity. It was launched in November 2004 and is coordinated by IBM with client software available for Windows, Linux, Mac OS X and Android OSes.

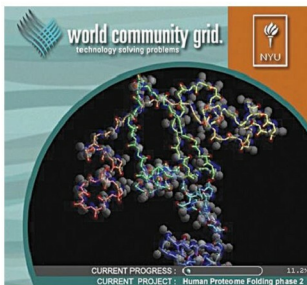
In 2003, IBM and other research participants sponsored Smallpox Research Grid Project to fast-track the discovery of a cure for smallpox. The study used a huge distributed computing grid to analyse compounds' effectiveness against smallpox. The project enabled scientists to observe and filter 35 million potential drug molecules against several smallpox proteins to shortlist good candidates for developing into smallpox treatments.

In the first 72 hours, 100,000 results were returned. By the end of the project, 44 strong treatment

candidates were identified. Based on the success of the smallpox study, IBM announced the birth of WCG with the objective of creating a technical environment where other humanitarian research could thrive.



Folding@home project involves studying proteins (Image courtesy: <http://sinoplife.ru/en/projects>)



By October 2014, WCG project had more than 60,000 active registered applicants (Image courtesy: <http://scistarter.com>)

In the beginning WCG only supported Windows, using the proprietary Grid MP software from United Devices, which powered *grid.org* distributed computing projects. Popularity for Linux support led to the addition in November 2005 of open source BOINC grid technology, which powers projects such as SETI@home and Climateprediction. In 2007, WCG migrated from Grid MP to BOINC for all of its supported platforms.

The project uses idle time of Internet-connected computers around the world to perform research calculations and analyse aspects of the human genome, HIV, dengue, muscular dystrophy, cancer, influenza, Ebola, virtual screening, rice crop yields and clean energy. Users install WCG client software onto their computers. The software quietly works in the background, using spare system resources to process work for WCG.

When a task or work unit is completed, the client software sends it back to WCG over the Internet and downloads a new work unit. To ensure accuracy, WCG servers send out multiple copies of each work unit. Thereafter, on receipt of results, these are collected and validated against each other.

Applicants can select the graphics output by the current work unit as a screensaver. Public computing grids such as SETI@home and Folding@home are dedicated to a single project, while WCG caters to multiple humanitarian projects under a single banner. Users are included in a sub-set of projects by default, but can opt out of projects when they want. By

October 2014, the organisation had partnered with as many as 466 other companies and organisations to aid in its work, and had more than 60,000 active registered applicants.

Putting yourself on the grid computing map

Today, there are a number of such active grid computing projects. However, several of these projects are not on a continuous basis, that is, once the respective objectives are